



## Group 7

# Hierarchical Clustering

*Building a Family Tree for Your Data*

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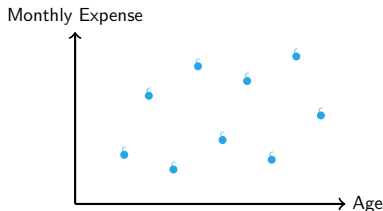
# Outline

- 1 Introduction - Clustering Basics
- 2 Transition to Hierarchical Clustering
- 3 Hierarchical Clustering - Introduction
- 4 Types of Hierarchical Clustering
  - Agglomerative Clustering
  - Divisive Clustering
- 5 Linkage Criteria

# A Problem to Think About

## Example (Superstore Customer Problem)

A large superstore wants to understand their 10,000 customers better. They have data about each customer:

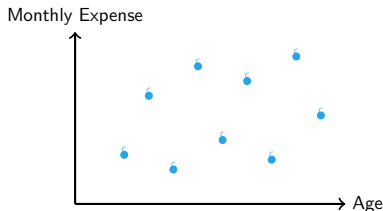


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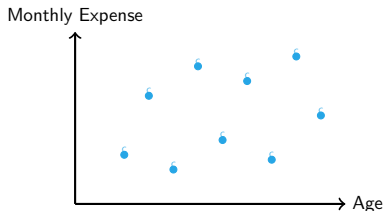


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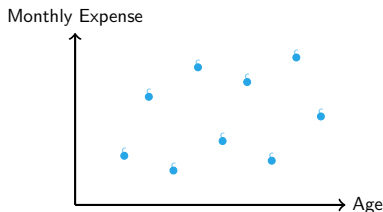


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## The Question

**How can they group these customers to create targeted offers for each group?**

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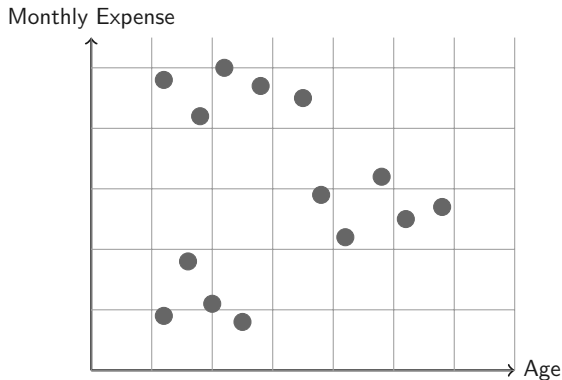
## Think

Manually grouping 10,000 customers? **Impossible!**



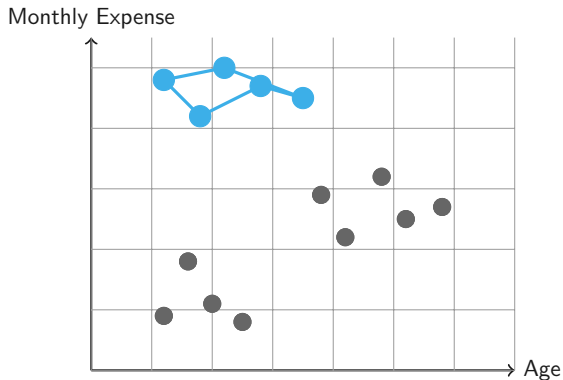
# Solution: Clustering!

- **Clustering** automatically groups similar customers



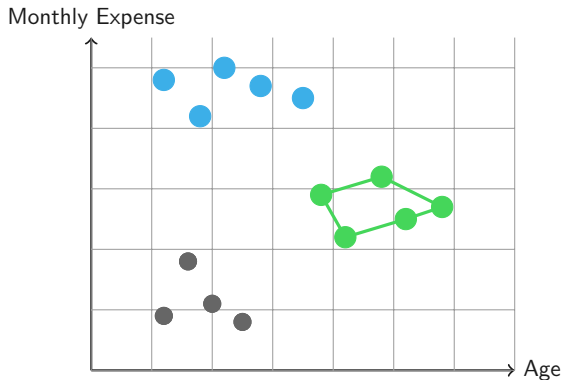
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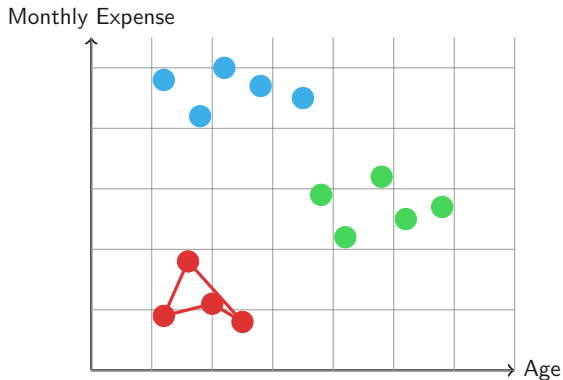
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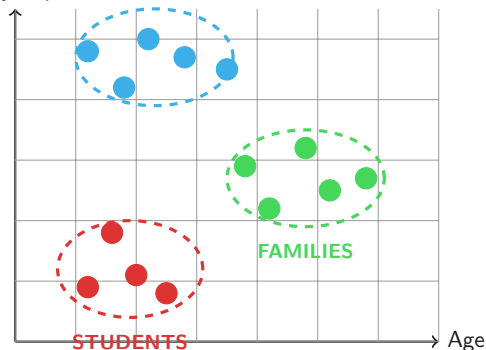
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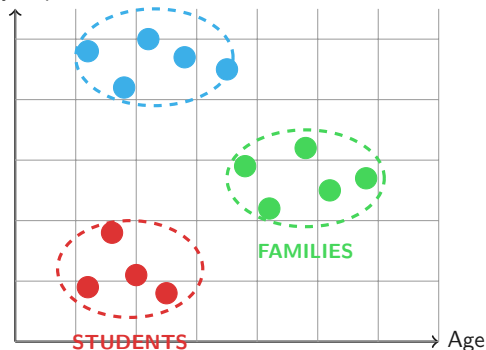
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- Boundaries define each cluster with **meaningful names**

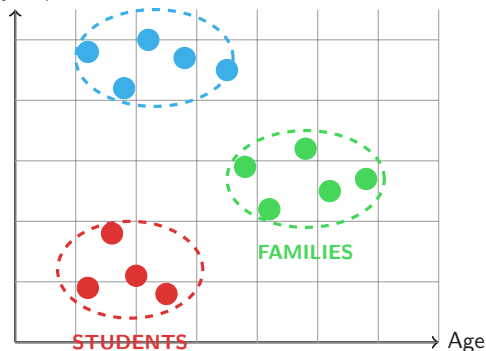
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# Solution: Clustering!

- **Clustering** automatically groups similar customers
- Each group = **Cluster**
- Boundaries define each cluster with **meaningful names**
- Customers in same cluster have similar:
  - Age
  - Spending habits
  - Preferences

Monthly Expense **PREMIUM**

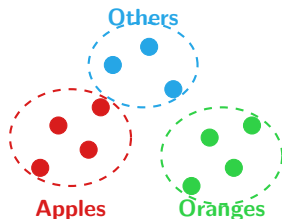


**Each cluster contains customers with similar characteristics!**

# What is Clustering?

## Definition

**Clustering** is the process of **dividing the datasets** into groups, consisting of **similar data-points**.



## Analogy

Like sorting a fruit basket: all **apples** go in one bowl, all **oranges** in another.



**But...**

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Regular clustering gives us **groups**...

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But it doesn't tell us **how groups are related** to each other!

**We need something more...**

# Here Comes...



## Hierarchical Clustering!

# DONE!

## What we covered?

Real-life Problem Example

Clustering Fundamentals

The Need of Hierarchical Clustering

Over to **Speaker 2**

# Definition of Hierarchical Clustering

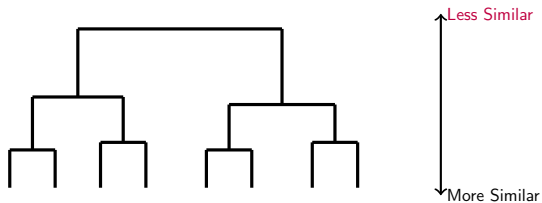
# Definition of Hierarchical Clustering

## Hierarchical Clustering

**Hierarchical cluster analysis** or **HCA** is a method of cluster analysis that builds a **tree structure** (*dendrogram*) from data similarities.



# Dendrogram



# Types of Hierarchical Clustering

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## Agglomerative

### Bottom-Up Approach

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## Divisive

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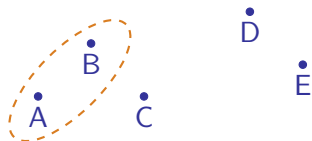
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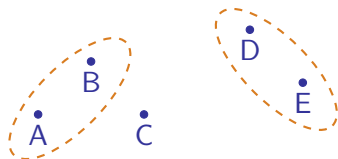
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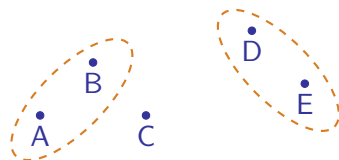




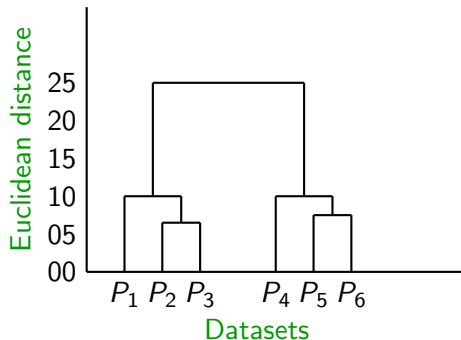
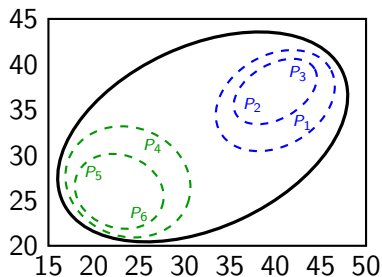
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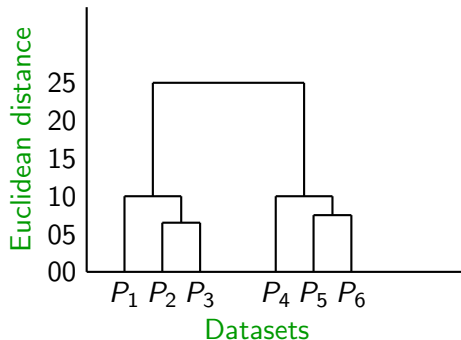
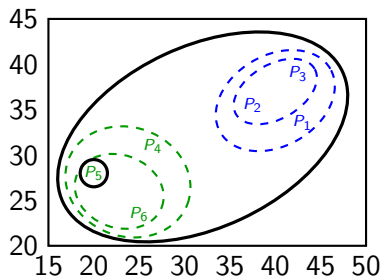
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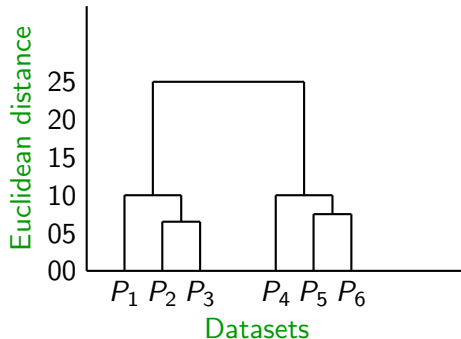
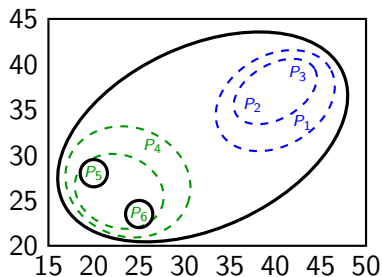
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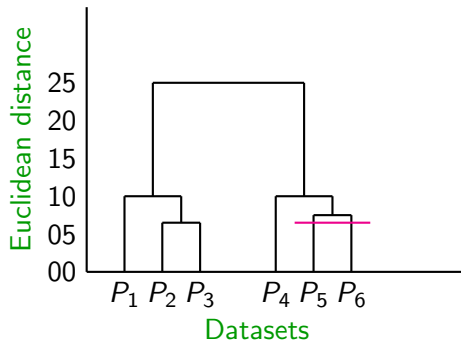
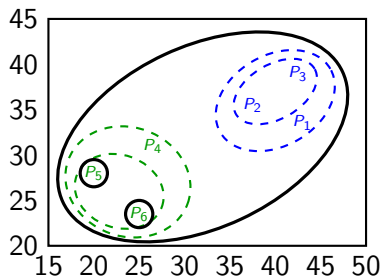
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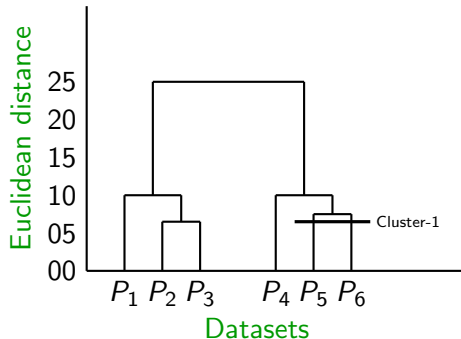
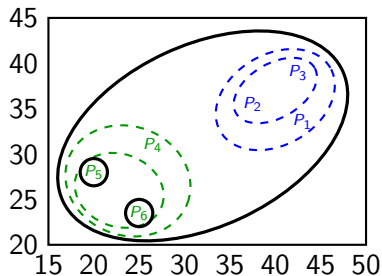
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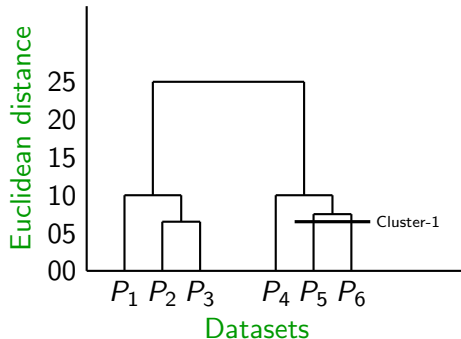
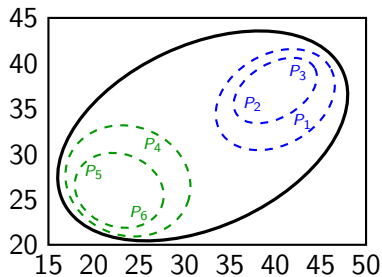
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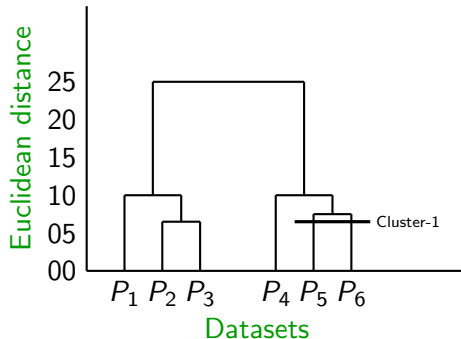
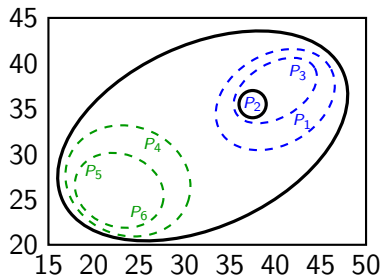
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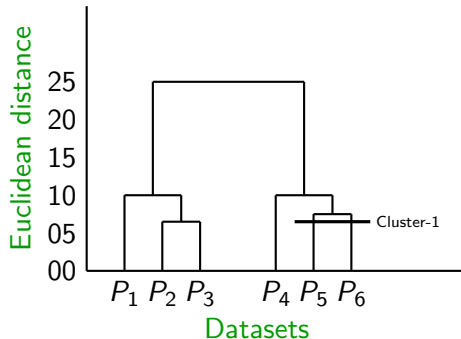
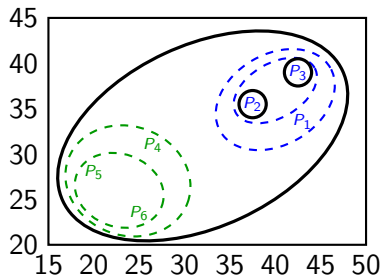


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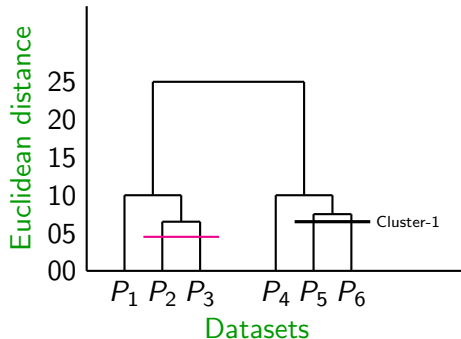
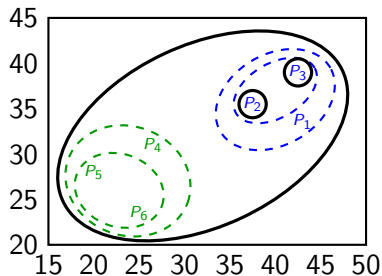




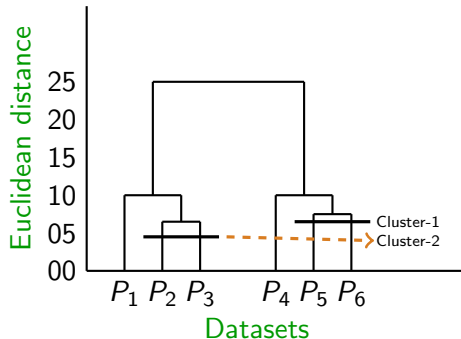
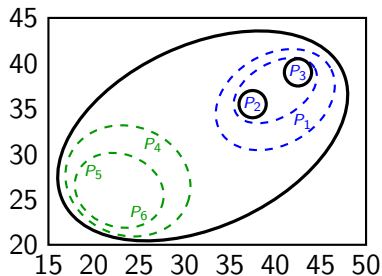
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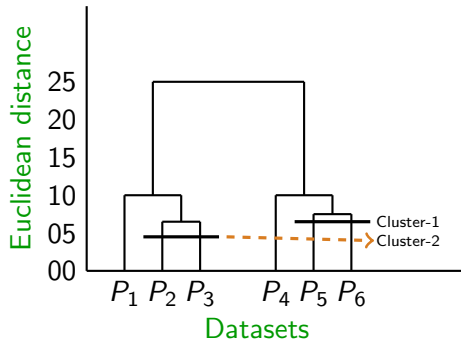
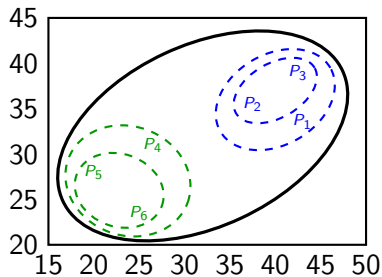
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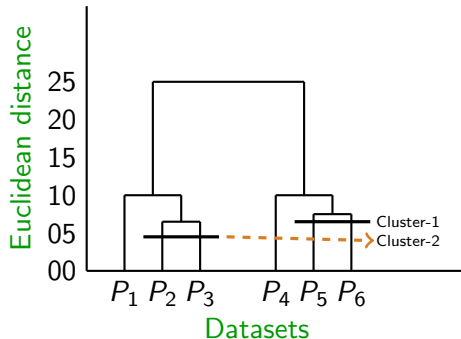
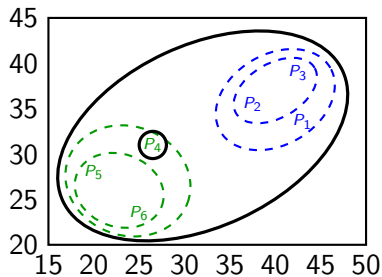
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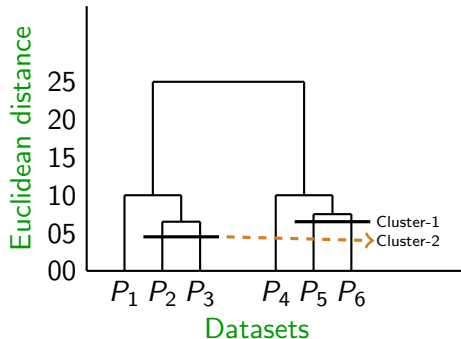
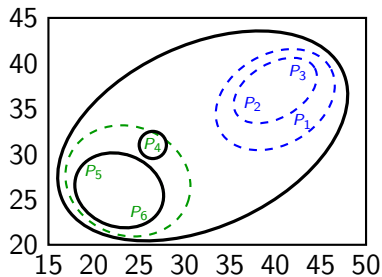
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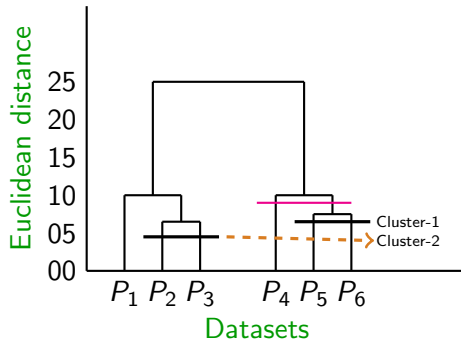
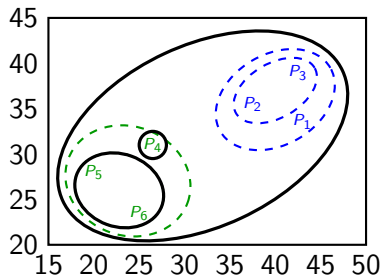
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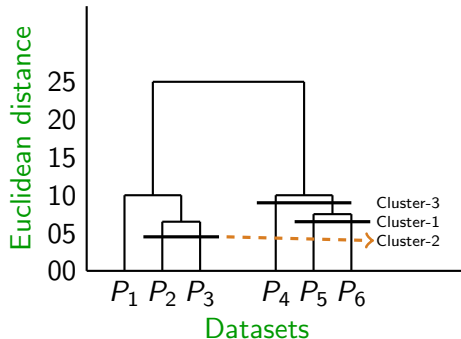
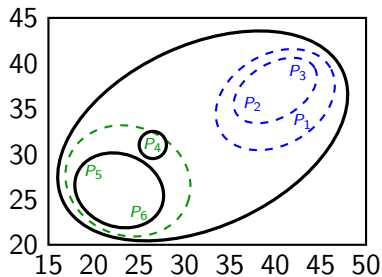
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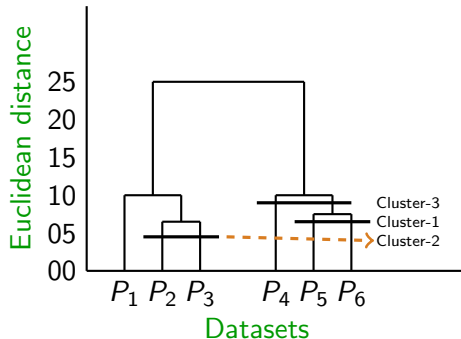
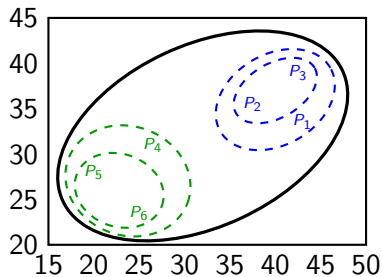


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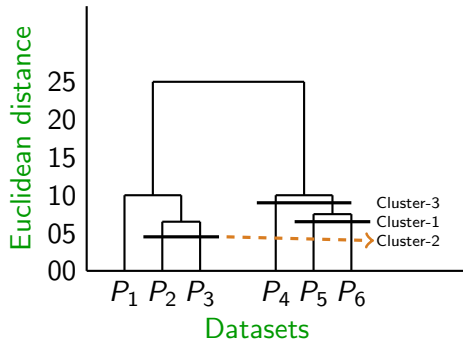
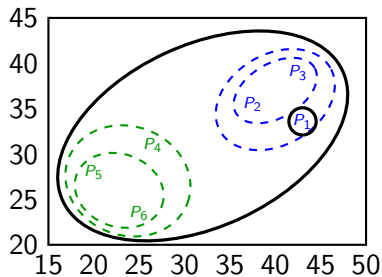




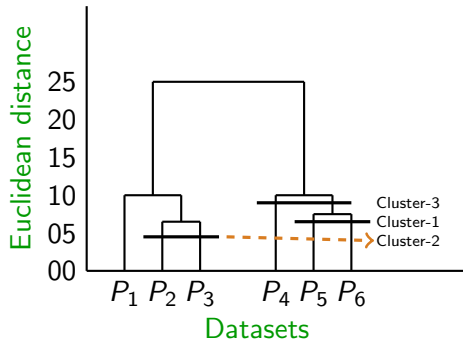
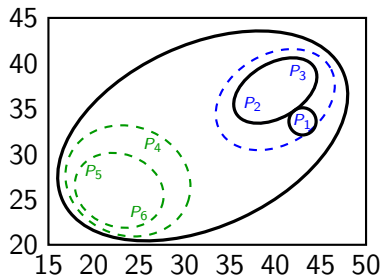
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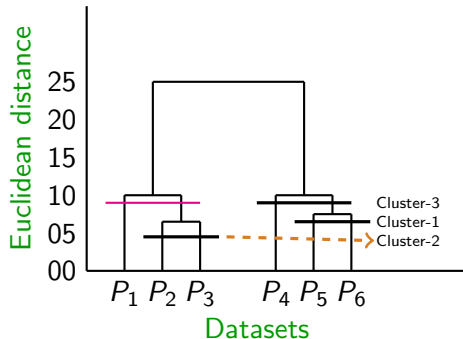
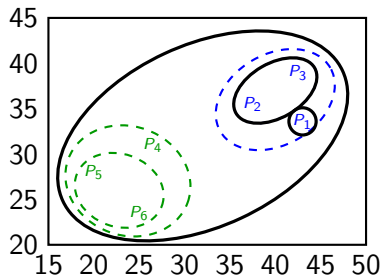
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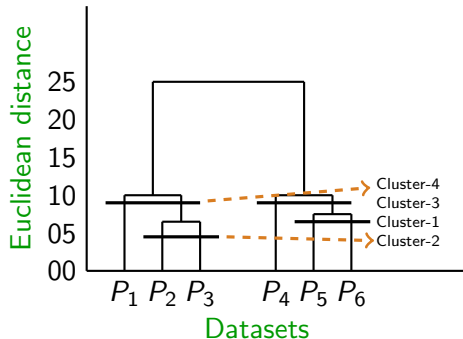
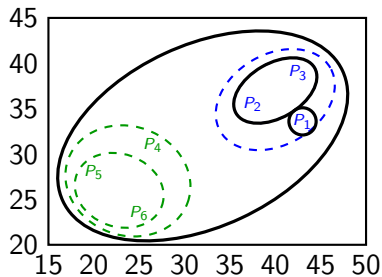
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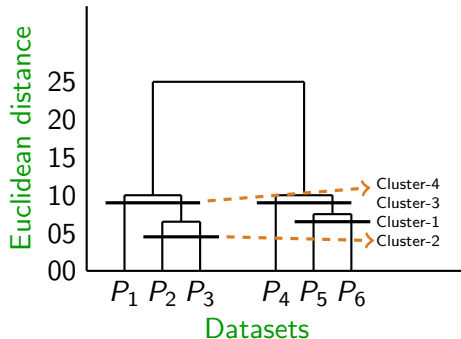
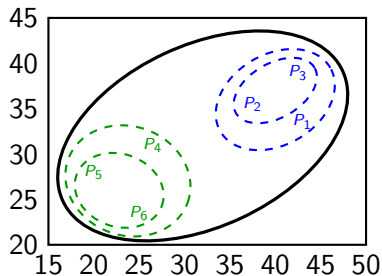
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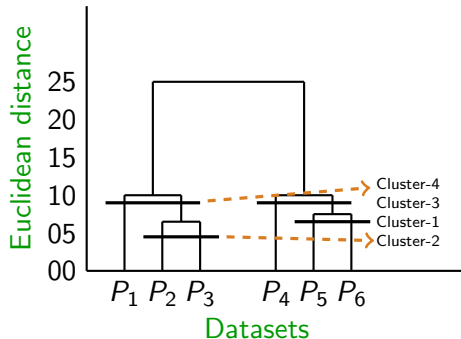
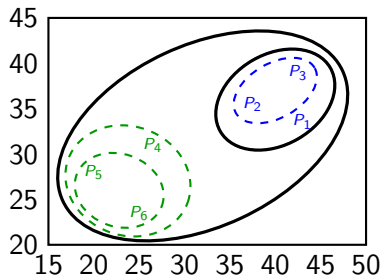
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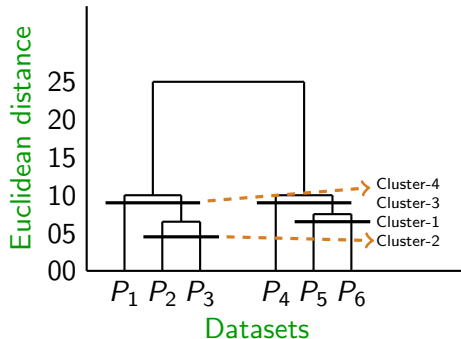
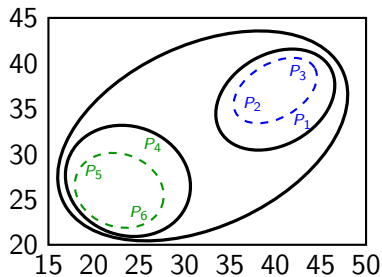
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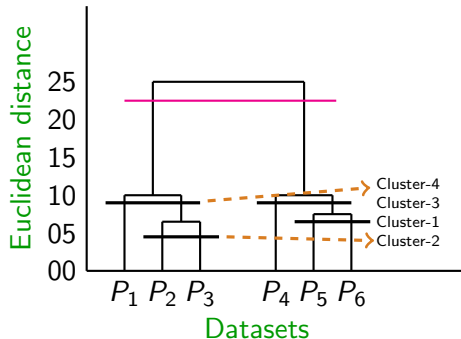
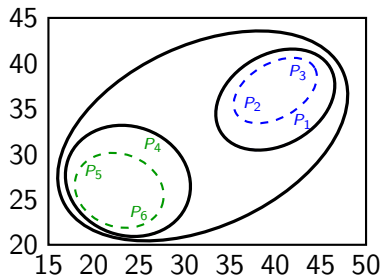


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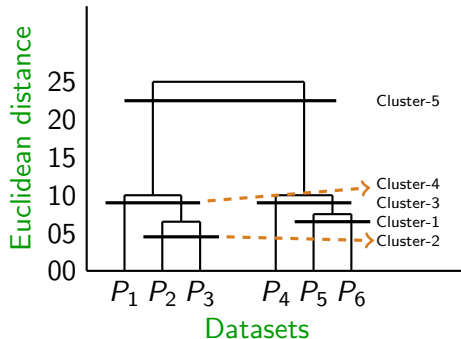
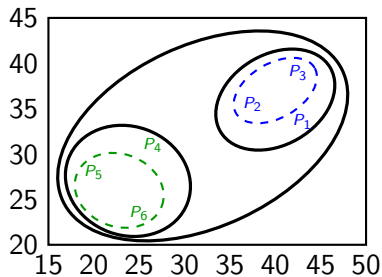




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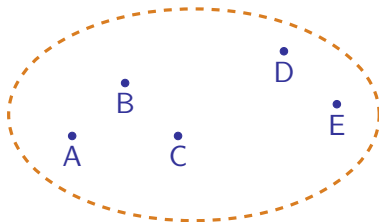
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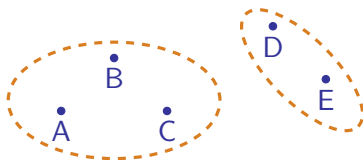
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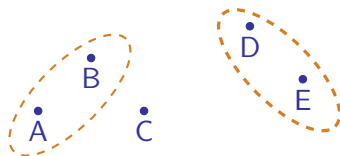
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- 2 **Split** a cluster into two smaller ones.



# Divisive Clustering

## Divisive (Top-Down)

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- 2 **Split** a cluster into two smaller ones.
- 3 Repeat until each point is its own cluster.



and so on...

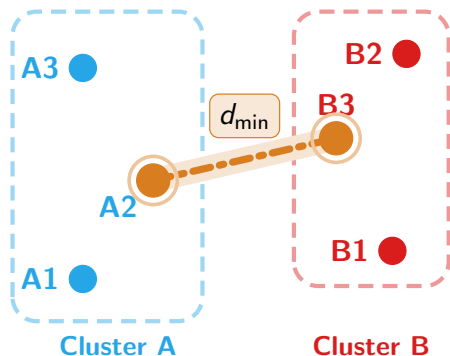
Which is more common?

**Agglomerative** is more popular because it's computationally simpler.

# How Do We Define “Closeness”? (Linkage Methods)

- **Single Linkage (Min):**

- Closest pair of points.
- Can lead to “chaining” (long, thin clusters).





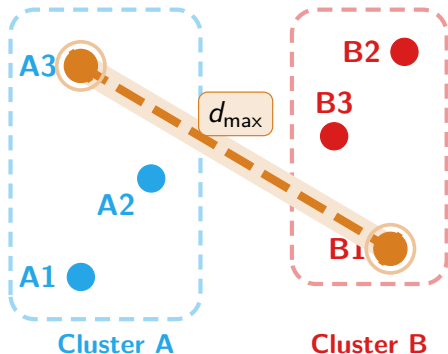
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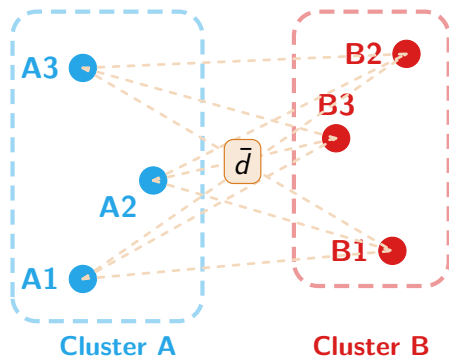
- **Complete Linkage (Max):**

- Furthest pair of points.
- *Produces compact, spherical clusters.*



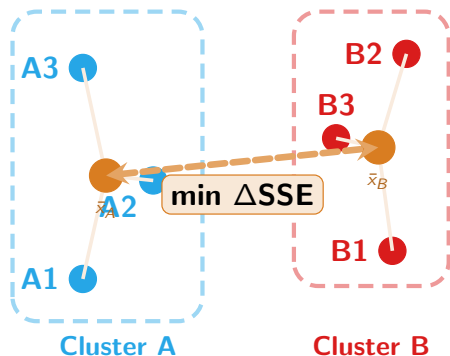
# How Do We Define “Closeness”? (Linkage Methods)

- **Single Linkage (Min):**
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  - *A robust “middle ground” approach.*
- **Ward’s Method:**
  - Minimizes within-cluster variance.
  - *Most popular for balanced data.*



# Single Linkage Clustering

**Single linkage distance:**

$$D(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$

**Distance Matrix**

|   | A  | B | C | D  |
|---|----|---|---|----|
| A | 0  | 2 | 6 | 10 |
| B | 2  | 0 | 5 | 9  |
| C | 6  | 5 | 0 | 4  |
| D | 10 | 9 | 4 | 0  |

**Dendrogram**

A                  B                  C                  D

# Single Linkage Clustering

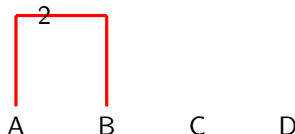
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**Dendrogram**



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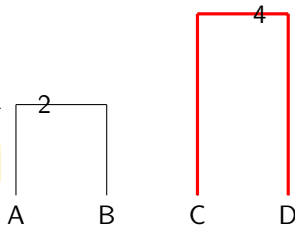
**Single linkage distance:**

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**Distance Matrix**

|           | <i>AB</i> | <i>C</i>         | <i>D</i>          |
|-----------|-----------|------------------|-------------------|
| <i>AB</i> | 0         | $\min(6, 5) = 5$ | $\min(10, 9) = 9$ |
| <i>C</i>  | 5         | 0                | 4                 |
| <i>D</i>  | 9         | 4                | 0                 |

**Dendrogram**



# Single Linkage Clustering

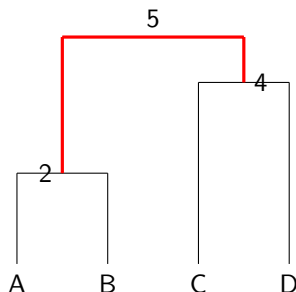
**Single linkage distance:**

$$D(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$

**Distance Matrix**

|           | <i>AB</i> | <i>CD</i>        |
|-----------|-----------|------------------|
| <i>AB</i> | 0         | $\min(5, 9) = 5$ |
| <i>CD</i> | 5         | 0                |

**Dendrogram**



# Complete Linkage Clustering

**Complete linkage distance:**

$$D(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$

**Distance Matrix**

|   | A  | B | C | D  |
|---|----|---|---|----|
| A | 0  | 2 | 6 | 10 |
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**Dendrogram**

A                  B                  C                  D



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**Dendrogram**



# Complete Linkage Clustering

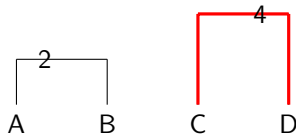
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**Dendrogram**



# Complete Linkage Clustering

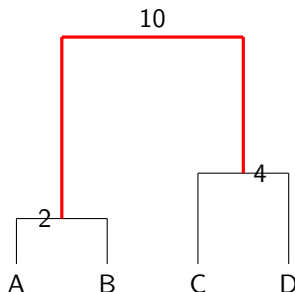
**Complete linkage distance:**

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**Distance Matrix**

|           | <i>AB</i> | <i>CD</i>          |
|-----------|-----------|--------------------|
| <i>AB</i> | 0         | $\max(6, 10) = 10$ |
| <i>CD</i> | 10        | 0                  |

**Dendrogram**



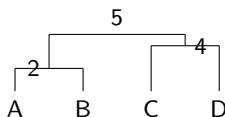
# Comparison: Single vs Complete Linkage

## Original Distance Matrix

|   | A  | B | C | D  |
|---|----|---|---|----|
| A | 0  | 2 | 6 | 10 |
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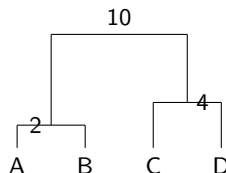
### Single Linkage

$$D(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$



### Complete Linkage

$$D(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$



*Same initial merges (A-B at 2, C-D at 4), but final merge height: 5 (single) vs. 10 (complete).*

# Average Linkage Clustering

**Average linkage distance:**

$$D(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d(x, y)$$

**Distance Matrix**

|   | A  | B | C | D  |
|---|----|---|---|----|
| A | 0  | 2 | 6 | 10 |
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**Dendrogram**

A      B      C      D

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**Dendrogram**



# Average Linkage Clustering

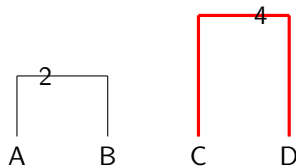
**Average linkage distance:**

$$D(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d(x, y)$$

**Distance Matrix**

|    | AB  | C                     | D                      |
|----|-----|-----------------------|------------------------|
| AB | 0   | $\frac{6+5}{2} = 5.5$ | $\frac{10+9}{2} = 9.5$ |
| C  | 5.5 | 0                     | 4                      |
| D  | 9.5 | 4                     | 0                      |

**Dendrogram**



# Average Linkage Clustering

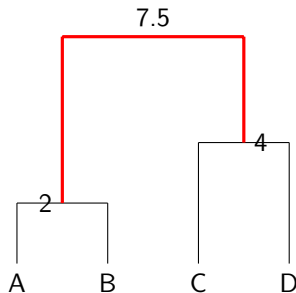
**Average linkage distance:**

$$D(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d(x, y)$$

**Distance Matrix**

|    | AB  | CD                         |
|----|-----|----------------------------|
| AB | 0   | $\frac{6+10+5+9}{4} = 7.5$ |
| CD | 7.5 | 0                          |

**Dendrogram**





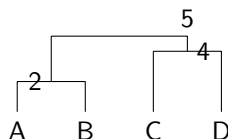
# Comparison of Linkage Methods

**Distance Matrix**

|   | A  | B | C | D  |
|---|----|---|---|----|
| A | 0  | 2 | 6 | 10 |
| B | 2  | 0 | 5 | 9  |
| C | 6  | 5 | 0 | 4  |
| D | 10 | 9 | 4 | 0  |

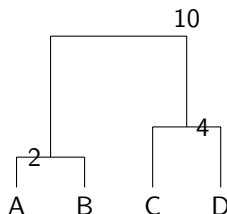
## Single

$$D_{\min} = \min d(x, y)$$



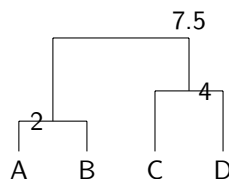
## Complete

$$D_{\max} = \max d(x, y)$$



## Average

$$D_{\text{avg}} = \frac{1}{n_i n_j} \sum d$$



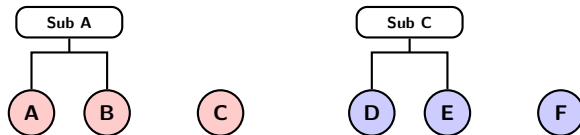
*Final merge heights: single 5, complete 10, average 7.5.*

# The Dendrogram: A Visual Hierarchy



Leaves: Individual data points.

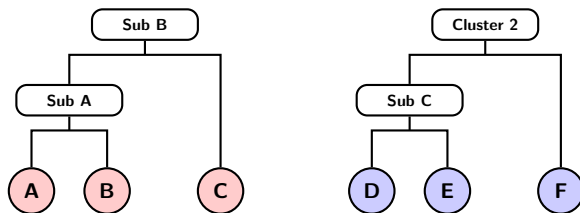
# The Dendrogram: A Visual Hierarchy



**Leaves:** Individual data points.

**Branches:** Groups formed as we merge similar data.

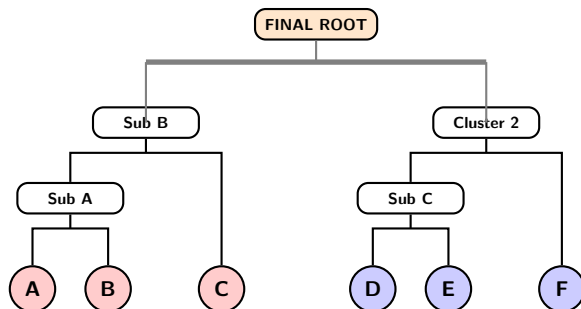
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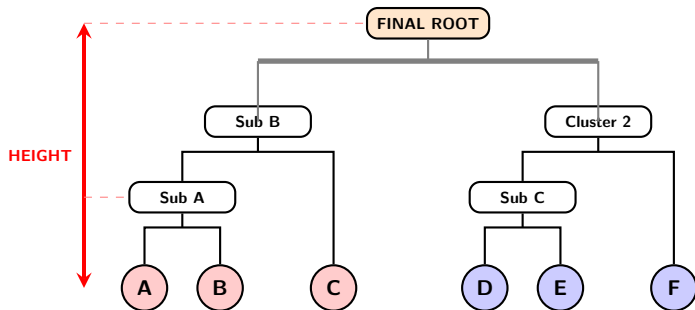


**Leaves:** Individual data points.

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**Root:** The final connection where everything is one.

# The Dendrogram: A Visual Hierarchy



**Leaves:** Individual data points.

**Branches:** Groups formed as we merge similar data.

**Root:** The final connection where everything is one.

**Height:** The distance at which clusters merge.

# Conclusion

## Let's recap the essential concepts:

- Hierarchical clustering builds a **tree of clusters**.



Thank You!

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- Hierarchical clustering builds a **tree of clusters**.
- Two main types: **Agglomerative** (bottom-up) and **Divisive** (top-down).
- **Linkage criteria** define how we measure cluster distance.
- The **dendrogram** is a powerful visualization tool.
- Widely used in bioinformatics (e.g., phylogenetics), finance (e.g., portfolio analysis), and marketing (e.g., customer segmentation).

